

# On Ramanujan Challenge Problem 3.2: The Apéry GCD Conjecture

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## Abstract

We study  $G_n = \gcd(d_n a_n, d_n b_n)$  for the Apéry sequences  $(a_n), (b_n)$  with  $d_n = \text{lcm}(1, \dots, n)^3$ . We prove *unconditionally* that  $G_n = e^{o(n)}$  for a set of positive integers of natural density 1. The key ingredient is a new bound  $Z(p) = o(p)$  on the zero-count function  $Z(p) = \#\{j < p : p \mid b_j\}$ , derived from the Apéry recurrence via a gap polynomial argument. Computation for all 9,590 primes  $5 \leq p \leq 10^5$  gives  $Z(p) \leq 10$  with mean  $\approx 1$ . Under the stronger arithmetic hypothesis that  $Z(p) = O(1)$  on average (Hypothesis Z), we prove  $\log G_n = O(\sqrt{n})$  for all  $n$ .

## 1 Setup

The Apéry sequences satisfy the three-term recurrence

$$(n+1)^3 u_{n+1} = (34n^3 + 51n^2 + 27n + 5) u_n - n^3 u_{n-1}, \quad n \geq 1, \quad (1)$$

with  $(a_0, a_1) = (0, 6)$  and  $(b_0, b_1) = (1, 5)$ . Here  $b_n = \sum_{k=0}^n \binom{n}{k}^2 \binom{n+k}{k}^2 \in \mathbb{Z}$  is the  $n$ -th Apéry number. Set  $d_n = \text{lcm}(1, \dots, n)^3$ ; Apéry [2] showed  $d_n a_n, d_n b_n \in \mathbb{Z}$ . Define  $G_n = \gcd(d_n a_n, d_n b_n)$ . Write  $P(n) = 34n^3 + 51n^2 + 27n + 5$  for the middle coefficient.

## 2 The Wronskian identity

**Lemma 1.** *For all  $n \geq 1$ ,  $W_n := a_n b_{n-1} - a_{n-1} b_n = 6/n^3$ .*

*Proof.* The Casorati determinant satisfies  $W_{n+1}/W_n = n^3/(n+1)^3$  from (1). Since  $W_1 = a_1 b_0 - a_0 b_1 = 6$ , telescoping gives  $W_n = 6/n^3$ .  $\square$

**Corollary 2.** *For every prime  $p \geq 5$ ,*

$$v_p(G_n) \leq 3 \lfloor \log_p n \rfloor. \quad (2)$$

*Proof.*  $G_n$  divides  $d_n(a_n b_{n-1} - a_{n-1} b_n) = 6d_n/n^3$ . For  $p \geq 5$ ,  $v_p(6) = 0$  and  $v_p(d_n/n^3) = 3 \lfloor \log_p n \rfloor - 3v_p(n) \leq 3 \lfloor \log_p n \rfloor$ .  $\square$

## 3 Small primes: unconditional bound

**Proposition 3.**  $\sum_{p \leq \sqrt{n}} v_p(G_n) \log p = O(\sqrt{n})$ .

*Proof.* By (2),  $v_p(G_n) \log p \leq 3 \log n$  for every prime  $p \geq 5$ . Primes  $p = 2, 3$  contribute at most  $O(\log n)$  each. There are  $O(\sqrt{n}/\log n)$  primes up to  $\sqrt{n}$ , so the sum is at most  $3 \log n \cdot O(\sqrt{n}/\log n) = O(\sqrt{n})$ .  $\square$

## 4 The multiplicative Lucas congruence

**Theorem 4** (Gessel [6]). *For every prime  $p$  and  $n = \sum_{i=0}^s n_i p^i$  in base  $p$ ,*

$$b_n \equiv \prod_{i=0}^s b_{n_i} \pmod{p}. \quad (3)$$

This follows from the combinatorial identity  $b_n = \sum_k \binom{n}{k}^2 \binom{n+k}{k}^2$  and termwise application of Lucas's congruence to each binomial coefficient. See also Mimura [10] and McIntosh [9].

## 5 The zero-count function

**Definition 5.** For a prime  $p \geq 5$ , define  $Z(p) = \#\{j \in \{0, \dots, p-1\} : b_j \equiv 0 \pmod{p}\}$ .

### 5.1 Unconditional bound

**Lemma 6** (No consecutive zeros). *For every prime  $p \geq 5$  and every  $j \in \{0, \dots, p-2\}$ ,  $b_j$  and  $b_{j+1}$  cannot both vanish modulo  $p$ .*

*Proof.* Suppose  $b_m \equiv b_{m+1} \equiv 0 \pmod{p}$  with  $0 \leq m \leq p-2$ . The recurrence (1) at  $n = m$  gives  $(m+1)^3 b_{m+1} = P(m) b_m - m^3 b_{m-1}$ , so  $m^3 b_{m-1} \equiv 0 \pmod{p}$ . Since  $1 \leq m \leq p-2$  implies  $p \nmid m$ , we get  $b_{m-1} \equiv 0$ . Iterating gives  $b_0 \equiv 0 \pmod{p}$ , contradicting  $b_0 = 1$ .  $\square$

**Lemma 7** (Gap polynomial). *Fix  $h \geq 2$ . Suppose  $b_m \equiv 0 \pmod{p}$  and  $m+h \leq p-1$ . Then*

$$b_{m+h} \equiv C_h(m) b_{m+1} \pmod{p}, \quad (4)$$

where  $C_h(m)$  is a rational function of  $m$  whose numerator (after clearing the denominator  $\prod_{j=1}^{h-1} (m+j+1)^3$ ) is a polynomial of degree exactly  $3(h-1)$  with positive leading coefficient.

In particular, if  $b_m \equiv b_{m+h} \equiv 0 \pmod{p}$ , then by Lemma 6  $b_{m+1} \not\equiv 0$ , so  $C_h(m) \equiv 0 \pmod{p}$ , and  $m$  is constrained to at most  $3(h-1)$  residue classes modulo  $p$ , provided  $C_h$  is not identically zero over  $\mathbb{F}_p$ .

*Proof.* We iterate the recurrence from the initial condition  $(b_m, b_{m+1}) = (0, b_{m+1})$ . At step  $n = m+1$ :  $b_{m+2} = P(m+1) b_{m+1} / (m+2)^3$ , since  $b_m = 0$  kills the second term. At step  $n = m+j$  for  $j \geq 2$ :  $b_{m+j+1}$  is a  $\mathbb{Q}(m)$ -linear combination of  $b_{m+j}$  and  $b_{m+j-1}$ , both already rational multiples of  $b_{m+1}$ . The leading coefficient of  $P(n)$  is 34 and the subtracted term has leading coefficient 1, so at each step the degree in  $m$  increases by 3 and the leading coefficient remains positive. After  $h-1$  steps the numerator has degree  $3(h-1)$ .

For  $h = 2$ :  $C_2(m) = P(m+1)/(m+2)^3$ , with numerator  $P(m+1) = (2m+3)(17m^2+51m+39)$ , a cubic with leading coefficient 34.  $\square$

**Lemma 8** (Nonvanishing over  $\mathbb{F}_p$ ). *For every prime  $p \geq 7$  and every  $2 \leq h \leq p$ , the numerator of  $C_h$  is not the zero polynomial modulo  $p$ .*

*Proof.* Write  $N_h$  for the numerator of  $C_h$  (a polynomial of degree  $3(h-1)$ ), satisfying  $N_{h+1}(m) = P(m+h) N_h(m) - (m+h)^6 N_{h-1}(m)$ , with  $N_1 = 1$  and  $N_2 = P(m+1)$ .

*First evaluation.* Equation (4) at  $m = -1$  (where  $b_{-1} = 0$ ,  $b_0 = 1$ ) gives  $C_h(-1) = b_{h-1}$ , hence  $N_h(-1) = b_{h-1} \cdot ((h-1)!)^3$ .

*Second evaluation.* Setting  $m = -2$  in the recurrence gives  $N_{h+1}(-2) = P(h-2)N_h(-2) - (h-2)^6 N_{h-1}(-2)$ , with  $N_1(-2) = 1$  and  $N_2(-2) = P(-1) = -5$ . By induction using the Apéry recurrence at  $n = h - 2$ :

$$N_h(-2) = -5 b_{h-2} ((h-2)!)^3 \quad (h \geq 2). \quad (5)$$

*Conclusion.* If  $N_h \equiv 0 \pmod{p}$  as a polynomial ( $2 \leq h \leq p$ ,  $p \geq 7$ ), then  $N_h(-1) \equiv N_h(-2) \equiv 0$ , forcing  $b_{h-1} \equiv b_{h-2} \equiv 0 \pmod{p}$  (since  $p \nmid 5$  and  $p \nmid (h-1)!, (h-2)!$ ). These are consecutive zeros, contradicting Lemma 6.  $\square$

**Proposition 9.** *For every prime  $p \geq 7$ ,*

$$Z(p) \leq \left\lfloor \frac{p}{H} \right\rfloor + \frac{3}{2}(H-1)(H-2) + 1, \quad (6)$$

where  $H$  is any integer with  $2 \leq H \leq p$ . Optimizing  $H = \lfloor (p/3)^{1/3} \rfloor + 1$  gives  $Z(p) \leq \frac{3^{4/3}}{2} p^{2/3} + O(p^{1/3}) = O(p^{2/3})$  unconditionally for all primes. In particular,  $Z(p) = o(p)$ .

*Proof.* Partition  $\{0, \dots, p-1\}$  into  $\lceil p/H \rceil$  consecutive blocks of length at most  $H$ . In each block, mark the first zero; all other zeros have gap  $h \in \{2, \dots, H-1\}$  to some earlier zero in the same block (gap 1 is impossible by Lemma 6). By Lemma 7, a gap- $h$  pair constrains  $m$  to at most  $3(h-1)$  residue classes mod  $p$  (the denominator factors  $(m+j+1)^3$  are nonzero since  $0 \leq m+j+1 \leq p-1$ ). By Lemma 8,  $C_h \not\equiv 0$  over  $\mathbb{F}_p$ , so the numerator roots are at most  $3(h-1)$ . Summing:

$$Z(p) \leq \left\lfloor \frac{p}{H} \right\rfloor + \sum_{h=2}^{H-1} 3(h-1) = \frac{p}{H} + \frac{3}{2}(H-1)(H-2) + O(1).$$

The minimum over real  $H > 0$  occurs at  $H = (p/3)^{1/3}$ , giving the bound. For  $p = 2, 3, 5$ :  $Z(p) = 0$  (since  $b_j$  for  $j < 5$  is coprime to  $2, 3, 5$ ).  $\square$

*Remark 10.* The leading coefficient of the numerator of  $C_h$  is  $U_{h-1}(17)$ , the Chebyshev polynomial of the second kind evaluated at 17, satisfying  $\ell_1 = 1$ ,  $\ell_{h+1} = 34\ell_h - \ell_{h-1}$ . Let  $N_h(m)$  denote the cleared numerator of  $C_h$ . The identity  $P(-u-1) = -P(u)$  (evident from  $P(n) = (2n+1)(17n^2 + 17n + 5)$ ) propagates through the recurrence to give  $N_h(-m-h-1) = (-1)^{h-1} N_h(m)$ , verified computationally for  $h \leq 200$ . For even  $h$ ,  $N_h$  is an *odd* polynomial in  $(m + (h+1)/2)$ , forcing the structural factor  $(2m+h+1)$ . At  $h = 2$ : factor  $(2m+3)$  in  $P(m+1) = (2m+3)(17m^2 + 51m + 39)$ . The computational data (Section 9) suggests  $Z(p) = O(1)$ , far stronger than  $O(p^{2/3})$ .

*Remark 11* (Content restriction). A stronger nonvanishing holds: if  $p \geq 7$  divides the content of  $N_h$  (the gcd of all its coefficients), then  $h \geq 2p$ . Indeed,  $N_h(-r) = (-1)^{r-1} b_{r-1} b_{h-r} ((r-1)!)^3 ((h-r)!)^3$  for  $1 \leq r \leq h$  (by induction from the Apéry recurrence); choosing  $r = 1, 2$  and using  $p \nmid (h-1)!, (h-2)!$ , 5 for  $h \leq p$  recovers Lemma 8, while the range  $p < h < 2p$  is excluded similarly via  $r = h - p + 1$ .

*Remark 12* (Resultant coprimality). The gap polynomials are pairwise coprime over  $\mathbb{Q}[m]$ . The resultants are extremely small:  $\text{Res}(N_2, N_3) = -5^6$ ,  $\text{Res}(N_3, N_4) = -2^{18} \cdot 5^6 \cdot 73^6$ ,  $\text{Res}(N_4, N_5) = 2^{36} \cdot 3^{18} \cdot 5^{12} \cdot 17^{12} \cdot 73^6$ . Consequently, for all primes  $p \geq 7$ , the root sets  $\{m : N_h(m) \equiv 0\}$  and  $\{m : N_k(m) \equiv 0\}$  are disjoint in  $\mathbb{F}_p$  for all but finitely many  $p$  (depending on  $h, k$ ). For  $N_2$  and  $N_3$ , disjointness holds for *every*  $p \geq 7$ . This structural independence supports the Poisson model of Section 9, and may be relevant for multi-gap improvements beyond  $O(p^{2/3})$ .

## 5.2 Computational evidence

**Proposition 13.** *For all 9,590 primes  $5 \leq p \leq 10^5$ :*

1.  $Z(p) \in \{0, 2, 4, 6, 8, 10\}$  for all but two primes; the two exceptions ( $p = 11$  and  $p = 3137$ ) have  $Z(p) = 1$  (non-ordinary primes, where  $p \mid a_p(f)$ ).  $\max Z(p) = 10$  at  $p = 88,609$ .
2. The exact histogram is:

| $Z(p)$                    | 0    | 1     | 2    | 4    | 6    | 8    | 10    |
|---------------------------|------|-------|------|------|------|------|-------|
| count                     | 5820 | 2     | 2926 | 722  | 106  | 13   | 1     |
| fraction                  | .607 | .0002 | .305 | .075 | .011 | .001 | .0001 |
| $\text{Poi}(\frac{1}{2})$ | .607 | —     | .303 | .076 | .013 | .002 | .0002 |

The mean  $Z(p) = 0.990$ , stable across all ranges.

3. The symmetry  $b_j \equiv b_{p-1-j} \pmod{p}$  holds for all tested primes: zeros come in palindromic pairs  $\{j, p-1-j\}$ , so  $Z(p)$  is even (except at non-ordinary primes). The pair count  $Z(p)/2$  fits  $\text{Poisson}(1/2)$  to within sampling noise (table above): the  $j$ -th pair appears independently with probability  $\approx 1/(2p)$ .

## 5.3 Hypothesis Z

**Hypothesis 14** (Hypothesis Z).  $\frac{1}{\pi(N)} \sum_{p \leq N} Z(p) = O(1)$  as  $N \rightarrow \infty$ .

*Remark 15.* The analogous problem for the  $\zeta(2)$  Apéry numbers  $A_n = \sum_k \binom{n}{k}^2 \binom{n+k}{k}$  has a fundamentally different structure. Stienstra–Beukers [11] showed that  $A_p$  is related to a CM modular form, yielding  $A_{(p-1)/2} \equiv 0 \pmod{p}$  for all  $p \equiv 3 \pmod{4}$  — a *positive density* of primes with guaranteed zeros. In contrast, the  $\zeta(3)$  Apéry numbers are governed by the non-CM newform  $f \in S_4(\Gamma_0(8))$  (LMFDB label 8.4.a.a), for which no such systematic zero-forcing occurs. This is a structural reason why Hypothesis Z should hold for the  $\zeta(3)$  sequence.

*Remark 16.* Even the single-row slice  $Z^*(p) := \mathbf{1}[p \mid b_{(p-1)/2}]$  is connected to the non-ordinary primes of  $f$  via the Beukers congruence  $b_{(p-1)/2} \equiv a_p(f) \pmod{p}$  [4, 1]. The density-zero conjecture for non-ordinary primes of non-CM weight-4 forms is itself open [7]. Thus Hypothesis Z contains a known open problem as a special case.

## 6 The denominator connection lemma

**Lemma 17.** *Let  $p \geq 7$  be prime with  $n/2 < p \leq n$ , and set  $r = n - p$ . Then  $v_p(G_n) = 0$  if and only if  $p \nmid b_r$ .*

*Proof.* Since  $p > n/2$ , the only multiple of  $p$  in  $\{1, \dots, n\}$  is  $p$  itself, so  $v_p(d_n) = 3$ .

**Step 1.**  $v_p(d_n b_n) = 3 + v_p(b_n) \geq 3$ , so  $v_p(G_n) = 0$  requires  $v_p(d_n a_n) = 0$ , i.e.,  $v_p(a_n) = -3$ .

**Step 2 (variation of parameters).** From Lemma 1, the Casorati identity telescopes to

$$\frac{a_n}{b_n} = \sum_{m=1}^n \frac{6}{m^3 b_m b_{m-1}}. \quad (7)$$

For  $p > n/2$ , the only  $m \in \{1, \dots, n\}$  with  $p \mid m$  is  $m = p$ .

**Step 3.** By Theorem 4,  $b_p \equiv b_0 b_1 = 5 \pmod{p}$  (since  $p$  has base- $p$  digits  $(0, 1)$ ), and by the symmetry  $b_j \equiv b_{p-1-j}$ ,  $b_{p-1} \equiv b_0 = 1 \pmod{p}$ . Since  $p \geq 7$ , both are nonzero mod  $p$ .

The  $m = p$  term in (7) has  $v_p = -3$ ; all other terms have  $v_p \geq 0$ . Therefore

$$p^3 \cdot \frac{a_n}{b_n} \equiv \frac{6}{b_p b_{p-1}} \equiv \frac{6}{5} \pmod{p}.$$

Using  $b_n \equiv 5 b_r \pmod{p}$  (Theorem 4, since  $n = 1 \cdot p + r$ ):

$$p^3 a_n \equiv \frac{6}{5} \cdot 5 b_r = 6 b_r \pmod{p}.$$

Thus  $v_p(a_n) = -3$  iff  $p \nmid b_r$ . □

**Verification.** For all  $n \leq 300$  and primes  $p \in (n/2, n]$  with  $p \geq 7$ :  $v_p(G_n) = 0 \iff p \nmid b_{n-p}$  confirmed in 2101/2101 cases (100%).

## 7 The unconditional density-one theorem

**Theorem 18.** For a set of positive integers  $n$  of natural density 1,  $\log G_n = O(\sqrt{n})$ , and in particular  $G_n = e^{o(n)}$ .

*Proof.* Fix  $N$  large and consider  $n \in (N, 2N]$ . Write  $B(n) = \#\{\text{bad primes } p > \sqrt{n} \text{ with } v_p(G_n) \geq 1\}$ . By Proposition 3, the small-prime contribution is  $O(\sqrt{n})$  for every  $n$ . Each bad prime contributes  $v_p(G_n) \log p \leq 3 \log n$ , so  $\log G_n \leq O(\sqrt{n}) + 3B(n) \log n$ . It suffices to show  $B(n) = o(n/\log n)$  for all but  $o(N)$  values of  $n \in (N, 2N]$ .

For  $p > \sqrt{n}$ ,  $n$  has two base- $p$  digits:  $n = qp + r$  with  $1 \leq q \leq \sqrt{2N}$  and  $0 \leq r < p$ . By Theorem 4,  $p \mid b_n$  iff  $p \mid b_q$  or  $r \in \mathcal{Z}_p$ , where  $\mathcal{Z}_p = \{j < p : p \mid b_j\}$ . A prime  $p$  can be bad for  $n$  only if one of these holds.

**Lower-digit incidence.** For fixed  $p \in (\sqrt{N}, 2N]$ , an interval of  $N$  consecutive integers contains at most  $N \cdot Z(p)/p + Z(p)$  values of  $n$  whose residue  $r = n \bmod p$  lies in  $\mathcal{Z}_p$ . Set  $\rho_p = Z(p)/p$ . The total lower-digit incidence is

$$R_N \leq \sum_{\sqrt{N} < p \leq 2N} (N\rho_p + Z(p)).$$

By Proposition 9,  $\rho_p \rightarrow 0$ . Since  $\pi(2N) = O(N/\log N)$ :

$$\sum_p \rho_p = o(\pi(2N)) = o(N/\log N), \quad \sum_p Z(p) = \sum_p p\rho_p \leq 2N \sum_p \rho_p = o(N^2/\log N).$$

Hence  $R_N = o(N^2/\log N)$ .

**Leading-digit incidence.** For fixed  $q$ ,  $p \mid b_q$  requires  $p$  to be a prime divisor of  $b_q$ , and  $b_q \leq (q+1) \cdot 64^q$  gives at most  $O(q)$  such primes. For each pair  $(q, p)$ , at most  $p$  integers in  $(N, 2N]$  have  $\lfloor n/p \rfloor = q$ . Hence the total leading-digit incidence is

$$L_N \leq \sum_{q=1}^{\lceil \sqrt{2N} \rceil} O(q) \cdot \frac{2N}{q+1} = O(N\sqrt{N}) = o(N^2/\log N).$$

**Conclusion.** By Markov's inequality applied to  $\sum_{n \in (N, 2N]} B(n) \leq R_N + L_N = o(N^2/\log N)$ : the number of  $n \in (N, 2N]$  with  $B(n) > n/\log^2 n$  is  $o(N)$ . A dyadic decomposition shows the exceptional set has density zero. For all remaining  $n$ ,  $\log G_n = O(\sqrt{n}) + 3B(n) \log n \leq O(\sqrt{n}) + O(n/\log n) = o(n)$ . □

## 8 Counting bad primes: the conditional result

**Theorem 19.** *Assume Hypothesis Z. Then  $\log G_n = O(\sqrt{n})$ , and in particular  $G_n = e^{o(n)}$ .*

*Proof.* As in Theorem 18, decompose  $\log G_n = \sum_p v_p(G_n) \log p$ . The small-prime contribution is  $O(\sqrt{n})$  unconditionally.

For primes  $p > \sqrt{n}$ :  $v_p(G_n) \leq 3$  and  $p$  is bad only if some base- $p$  digit of  $n$  lies in  $\mathcal{Z}_p$ . For  $p \in (n/2, n]$ :  $q = 1$ ,  $b_1 = 5$ , so  $p \nmid b_q$  for  $p \geq 7$ . By Lemma 17,  $p$  is bad iff  $r = n - p \in \mathcal{Z}_p$ . For  $p \in (\sqrt{n}, n/2]$ :  $p$  can be bad if  $q \in \mathcal{Z}_p$  or  $r \in \mathcal{Z}_p$ .

Under Hypothesis Z with average  $Z(p) \leq C$ : the count of bad primes satisfies

$$B(n) \leq \sum_{\substack{\sqrt{n} < p \leq n \\ p \text{ prime}}} \frac{2Z(p)}{p} \leq 2C \sum_{p > \sqrt{n}} \frac{1}{p} = O(1).$$

Each bad prime contributes  $v_p(G_n) \log p \leq 3 \log n$ . Hence  $\log G_n = O(\sqrt{n}) + O(\log n) = O(\sqrt{n})$ .  $\square$

**Corollary 20.** *The conjecture  $G_n = e^{o(n)}$  is equivalent to the average form of Hypothesis Z: if  $Z(p) \geq \varepsilon p$  for a positive proportion of primes, double-counting gives  $\log G_n \geq c\varepsilon n$  for a positive proportion of  $n$ .*

## 9 Computational verification

We computed  $G_n$  exactly for  $n = 1, \dots, 500$  using rational arithmetic.

| $n$ range | avg $\log(G_n)/n$ | max $\log(G_n)/n$ |
|-----------|-------------------|-------------------|
| 1–50      | 0.276             | 2.30              |
| 51–100    | 0.108             | 0.44              |
| 101–200   | 0.086             | 0.32              |
| 201–300   | 0.048             | 0.17              |
| 301–400   | 0.038             | 0.13              |
| 401–500   | 0.024             | 0.09              |

A power-law fit gives  $\log(G_n)/n \sim 4.5 n^{-0.85}$ , i.e.,  $\log G_n \sim 4.5 n^{0.15}$ , well within  $O(\sqrt{n})$ .

## 10 Remarks

1. The Beukers supercongruence  $b_{mp^r} \equiv b_{mp^{r-1}} \pmod{p^{3r}}$  [3] provides tower rigidity: for each fixed prime,  $v_p(b_n)$  is controlled by the base- $p$  digits.
2. For the gap-2 polynomial  $C_2(m) = P(m+1)$ , the factorization  $P(x) = (2x+1)(17x^2+17x+5)$  shows that gap-2 pairs can only occur at  $m = (p-3)/2$  (when  $2(m+1)+1 = 2m+3 \equiv 0$ ) or at roots of the quadratic  $17x^2+17x+5$  (discriminant  $-51$ ; exists over  $\mathbb{F}_p$  iff  $(\frac{-51}{p}) = 1$ ). Empirically, all observed gap-2 pairs arise from the linear factor at  $m = (p-3)/2$ .
3. For  $1 \leq j \leq p-2$ , the Apéry number  $b_j \bmod p$  admits a character-sum representation as a Greene [5] Gaussian hypergeometric  ${}_4F_3$  over  $\mathbb{F}_p$  with parameter  $\chi = \omega^j$  (Teichmüller character). However, the Weil bound controls the archimedean size  $|b_j|_\infty = O(p^{3/2})$  but does *not* bound  $Z(p)$ , since zeros of  $b_j$  modulo  $p$  are a  $p$ -adic event. The gap polynomial argument above is specifically adapted to this  $p$ -adic zero-count problem.

4. Malik–Straub [8] studied Lucas congruences and zero-divisibility for all sporadic Apéry-like sequences. Our  $Z(p)$  data for the  $\zeta(3)$  sequence agrees with their empirical findings.

## References

- [1] S. Ahlgren and K. Ono, *A Gaussian hypergeometric series evaluation and Apéry number congruences*, J. reine angew. Math. **518** (2000), 187–212.
- [2] R. Apéry, *Irrationalité de  $\zeta(2)$  et  $\zeta(3)$* , Astérisque **61** (1979), 11–13.
- [3] F. Beukers, *Some congruences for the Apéry numbers*, J. Number Theory **21** (1985), 141–155.
- [4] F. Beukers, *Another congruence for the Apéry numbers*, J. Number Theory **25** (1987), 201–210.
- [5] J. Greene, *Hypergeometric functions over finite fields*, Trans. Amer. Math. Soc. **301** (1987), 77–101.
- [6] I. Gessel, *Some congruences for Apéry numbers*, J. Number Theory **14** (1982), 362–368.
- [7] S. Jin, W. Ma, and K. Ono, *A note on non-ordinary primes*, Proc. Amer. Math. Soc. **144** (2016), 4591–4597.
- [8] A. Malik and A. Straub, *Divisibility properties of sporadic Apéry-like numbers*, Res. Number Theory **2** (2016), Art. 5.
- [9] R. J. McIntosh, *A generalization of a congruential property of Lucas*, Amer. Math. Monthly **99** (1992), 231–238.
- [10] Y. Mimura, *Congruence properties of Apéry numbers*, J. Number Theory **16** (1983), 138–146.
- [11] J. Stienstra and F. Beukers, *On the Picard–Fuchs equation and the formal Brauer group of certain elliptic K3 surfaces*, Math. Ann. **271** (1985), 269–304.